

JIAYI SHI

Business Intelligence Developer | Power BI Developer

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PROFESSIONAL SUMMARY

Business Intelligence Developer with 8+ years of experience designing, developing, and supporting BI and analytics solutions, with strong expertise in Power BI, SQL, DAX, Power Query, and dimensional data modeling. Experienced in translating business requirements into interactive dashboards, KPI frameworks, star-schema reporting models, and decision-support solutions for operational and leadership teams. Strong background in data transformation, data validation, performance optimization, stakeholder requirements gathering, and end-to-end BI development from initial business requirements through deployment and production support. Recipient of Freedom Mortgage's High Flyer award for outstanding BI contributions.

CORE TECHNICAL SKILLS

BI & Visualization: Power BI, DAX, Power Query, Power Pivot, Paginated Reports, Tableau, WebFocus

Data & Reporting: SQL, T-SQL, PL/SQL, SQL Server, Oracle, Data Validation, Data Transformation, Reporting Workflows

Data Modeling: Dimensional Modeling, Star Schema, KPI Design, Data Relationships, Reporting Models

Other: Python, Azure, Power Platform, Agile/Scrum, Requirements Gathering, Technical Documentation

PROFESSIONAL EXPERIENCE

Power BI Developer / BI Consultant | *Freelance, Remote*

Feb 2024 – Present

- Design and develop Power BI analytics solutions by translating stakeholder questions and reporting requirements into KPIs, data models, and interactive dashboard workflows.
- Transform, reconcile, and validate source data using SQL and Power Query, establishing analysis-ready datasets and reporting logic before visualization development.
- Build DAX measures, KPI frameworks, drill-through experiences, dynamic analysis views, and geospatial reporting to support executive-level monitoring and detailed investigation.
- Apply dimensional modeling and star-schema concepts to organize reporting data, simplify relationships, and improve analytical usability.
- Partner with stakeholders throughout the reporting lifecycle to clarify requirements, refine metrics, validate results, and adjust dashboard experiences based on business feedback.
- Review query, model, and visualization design to improve report performance, maintainability, and end-user usability.

Business Intelligence Developer | *Freedom Mortgage Inc., Cherry Hill, NJ*

Jan 2018 – Dec 2023

- Designed, developed, and supported enterprise BI reporting solutions that provided operational and leadership teams with visibility into business performance, trends, and reporting metrics.
- Partnered with business stakeholders to gather requirements, define KPI logic, identify reporting priorities, and translate business questions into dashboards and recurring reporting solutions.
- Used SQL and BI platforms to investigate source data, validate calculations, reconcile reporting results, and resolve data-quality or business-definition discrepancies prior to production release.
- Contributed to BI modernization and migration initiatives across WebFocus, Tableau, and Power BI-related reporting workflows, supporting the transition and maintenance of enterprise reporting solutions.
- Supported the full reporting lifecycle, including development, testing, deployment, user access, production troubleshooting, documentation, and ongoing enhancement.
- Collaborated with cross-functional development, business, and support teams in an Agile environment to coordinate reporting changes and production releases.
- Shared BI development practices and reporting knowledge with team members and received Freedom Mortgage's High Flyer award for outstanding contributions.

Data Analyst / Tableau Developer | *American Mountinvest Holding Company, Bala Cynwyd, PA*

Oct 2016 – Dec 2017

- Developed Tableau and WebFocus dashboards using SQL-based reporting data to provide business users with clearer visibility into operational trends and performance.
- Automated recurring reporting processes with SQL and standardized reporting logic, reducing manual effort and improving reporting consistency.
- Worked directly with users to understand analytical questions and refine dashboard metrics, filters, and views based on evolving business requirements.

Business Intelligence Developer | *TMG Health Inc., Jessup, PA*

Jul 2015 – Oct 2016

- Extracted, transformed, and prepared healthcare reporting data using SQL to support recurring BI deliverables and stakeholder analysis.
- Collaborated with development and UAT teams to validate reporting outputs, investigate data discrepancies, and support reliable production releases.

SELECTED POWER BI & ANALYTICS PROJECTS

U.S. PROCUREMENT ANALYTICS DASHBOARD

- Developed a Power BI procurement analytics solution to analyze public-sector spending, vendor concentration, agency activity, contract categories, and award timelines.
- Transformed procurement transaction data in Power Query and structured the reporting model around agency, recipient/vendor, category, contract, and time dimensions.
- Created DAX measures for total spend, agency/category performance, vendor concentration, and contract award cycle time.
- Built interactive filtering and drill-through analysis that allows users to move from executive-level spend trends to individual vendors and contracts.
- Framed the analysis around procurement decisions involving spend concentration, sourcing opportunities, vendor consolidation, and contract efficiency.

AUSTIN MUNICIPAL PROCUREMENT ANALYTICS DASHBOARD

- Developed a Power BI municipal procurement analytics solution to analyze City of Austin spending across vendors, commodities, contracts, and purchase orders.
- Transformed City of Austin open procurement data in Power Query and structured a star-schema reporting model around vendor, commodity, contract, and time dimensions.
- Created DAX measures for total spend, purchase order volume, vendor count, average purchase amount, and contracted versus non-contracted spend.
- Built executive, commodity, and contract analysis pages with interactive filtering and drill-through, framing the analysis around vendor concentration, commodity spending patterns, and contract efficiency.

NYC FHV / TAXI OPERATIONS ANALYTICS DASHBOARD

- Built an interactive Power BI solution to analyze trip demand by time, geography, airport activity, and operating conditions.
- Combined monthly trip data using Power Query and created DAX measures for trip volume, duration, distance, and peak-hour behavior.
- Designed dynamic filtering and comparison views to identify when and where demand is strongest and investigate the factors contributing to those patterns.

HEALTHCARE ANALYTICS DASHBOARD

- Developed a Power BI dashboard analyzing healthcare utilization, billing, admissions, length of stay, medical conditions, and operational patterns.
- Used Power Query and DAX to prepare reporting fields, define KPIs, and create interactive analytical views.
- Structured the dashboard around a KPI-to-detail workflow so users can move from high-level performance indicators to underlying patient and operational patterns.

INSURANCE FRAUD ANALYTICS DASHBOARD

- Developed an interactive Power BI solution to investigate fraud rates, claim exposure, high-risk patterns, and geographic concentrations.
- Created DAX measures for fraud rate, claim amount, and claim frequency, with drill-through analysis to investigate records behind high-level KPIs.
- Designed the analysis to identify claim patterns and geographic areas that may warrant additional investigation.

RETAIL SALES & CUSTOMER ANALYTICS DASHBOARD

- Built an executive Power BI reporting experience focused on sales performance, customer behavior, product performance, and profitability.
- Created KPI measures, Top-N analysis, dynamic filtering, tooltips, and drill-through interactions to identify performance changes and underlying business drivers.
- Applied dimensional reporting concepts to support a clear analytical path from executive KPIs to product and customer-level detail.

SUPPLY CHAIN & INVENTORY ANALYTICS DASHBOARD

- Developed a Power BI dashboard focused on inventory movement, fulfillment performance, supplier reliability, and operational bottlenecks.
- Created measures for Inventory Turnover, Fill Rate, and Supplier Performance with interactive filtering for deeper operational analysis.
- Designed the dashboard as a decision-support tool for identifying potential inventory and supplier issues requiring follow-up.

U.S. HOUSING MARKET ANALYTICS DASHBOARD

- Built an interactive Power BI solution to analyze U.S. housing price trends, inventory levels, days on market, and regional market patterns using Zillow Home Value Index (ZHVI) data.
- Transformed housing market data in Power Query and structured a star-schema reporting model around geography, property type, and calendar dimensions.
- Created DAX measures for median price, inventory, days on market, and YoY growth, with geospatial maps, state comparisons, and drill-through views to support market investigation.

EDUCATION & CERTIFICATIONS

M.S. Business Intelligence, Saint Joseph's University, Philadelphia, PA — 2015

B.S. General Management, Northwood University, Midland, MI — 2011

Certified BI Content Developer, SAS Institute Inc. | Tableau 10 Essential Certification